Goal: Optimal fixed stepsize for Gradient Descent under strong convexity.

Exercise 5.1

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is a L-smooth, convex function with minimizer x^*.

1. Show that for any $x \in \mathbb{R}^n$, we have

$$f(x) - f(x^*) \geq \frac{1}{2L}\|\nabla f(x)\|_2^2$$

   by minimizing the right hand side of the inequality

$$f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|_2^2.$$

   w.r.t. y.

2. Prove the co-coercivity property: For any $x, y \in \mathbb{R}^n$, we have

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|_2^2.$$

Exercise 5.2 (Optional)

Show that for any convex function $f : \mathbb{R}^n \to \mathbb{R}$, $(\nabla f(x) - \nabla f(y))^\top (x - y) \leq L\|x - y\|_2^2$ for all $x, y \in \mathbb{R}^n$ implies L-smoothness of f.

Exercise 5.3

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is a L-smooth, μ-convex function with (unique) minimizer x^*.

1. Show that the function $q(x) := f(x) - \frac{\mu}{2}\|x\|_2^2$ is convex and $(L - \mu)$-smooth. *Hint: Apply property from 5.2 to q.*

2. Use the co-coercivity property of previous exercises to the function q to show that the following property holds:

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \frac{\mu L}{\mu + L}\|x - y\|_2^2 + \frac{1}{\mu + L}\|\nabla f(x) - \nabla f(y)\|_2^2.$$

Exercise 5.4

Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is a μ-convex, L-smooth function with (unique) minimizer x^*. Use the co-coercivity property and the fact that $\nabla f(x^*) = 0$ to prove that the kth iterate of the steepest-descent method applied to f with steplength $\frac{2}{\mu+L}$ satisfies

$$\|x^k - x^*\|_2 \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k \|x^0 - x^*\|_2,$$

where $\kappa = \frac{L}{\mu}$. Finally, conclude about this convergence rate, compared to the one using fixed stepsize $1/L$.